

Multi-Seed Variance Evaluation

The AI Bidding War — outcome distributions on borderline scenarios

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Executive Summary

This report documents a multi-seed variance evaluation conducted in response to Phase 2 instructor feedback, which flagged the absence of a stochasticity study on borderline scenarios. Ten independent runs were executed on 2026-04-20 across the two scenarios that remain borderline under the Phase 3 model: Scenario 4 (near-ceiling) and Scenario 8 (exact-ceiling in the pinned harvest). Five runs per scenario — the top of the range the feedback requested.

Reliability is used here in a specific sense: the degree to which a single-harvest outcome is representative of the scenario’s behavior under repeated sampling.

Findings

- **Governance chain intact across all 10 runs.** Vendor isolation held, Strategic Audit fired before every Human-in-the-Loop decision, and no award was proposed above the \$15,000 ceiling. The buyer used the \$15,000 ceiling as a negotiation tactic in later rounds — repeatedly restating it to drive vendor prices down — so the ceiling held at the negotiation layer before deterministic enforcement was ever needed.
- **Negotiation outcomes varied; variance came from both sources.** Scenario 4 SUCCESS prices spanned \$13,500–\$15,000; Scenario 8 SUCCESS prices spanned \$13,500–\$14,200. Winner identity split between Vendor A (4) and Vendor C (4); Vendor B was never awarded. Attributing the spread solely to LLM stochasticity would be incomplete — the human gatekeeper made different decisions across runs on similar proposals, including one probe (“assess vendor a vs c; iso 27001 and p1?”) that rejected a \$15,000 proposal before approving a different \$15,000 award on the next turn.
- **Single-harvest ceiling claims are not reliable.** Scenario 8 closed at exactly \$15,000 in the pinned 2026-04-18 harvest; in 5 independent re-runs the SUCCESS price never reached \$15,000 (0/4 at ceiling). The pinned value falls outside the observed re-run range.

Architectural takeaway. The governance layer is deterministic; the negotiation layer is not. Variance at the negotiation layer — from LLM sampling and human judgment — is absorbed by the audit-before-approval and deterministic guardrail architecture, producing traceable outcomes under repeated sampling.

1. Phase 2 Feedback Context

The default buyer model was upgraded from *claude-sonnet-4-20250514* to *claude-sonnet-4-6* between Phase 2 and Phase 3 because the Phase 2 model is scheduled for deprecation in June 2026. The table below shows which scenarios remain borderline under the current model.

Scenario	Phase 2 pinned harvest	Phase 3 pinned harvest (2026-04-18)	Borderline in Phase 3?
4	\$15,000 (at ceiling)	\$14,750 (near-ceiling; \$250 below)	Yes
5	\$15,000 (at ceiling)	\$11,500 (\$3,500 below ceiling)	No
8	\$14,800 (near-ceiling)	\$15,000 (at exact ceiling)	Yes

2. Run Inventory

All ten runs were executed with identical inputs, varying only by LLM sampling and human operator decisions. Every run completed the full three-round negotiation before reaching its terminal decision.

#	Scn	Started (UTC)	Status	Final \$	Winner	At \$15k?	No contract ?
1	4	2026-04-20 22:31:31	SUCCESS	\$13,500	Vendor C	N	N
2	4	2026-04-20 22:22:45	NO_AWARD	—	—	—	Y
3	4	2026-04-20 22:15:06	SUCCESS	\$13,500	Vendor C	N	N
4	4	2026-04-20 22:05:39	SUCCESS	\$15,000	Vendor A	Y	N
5	4	2026-04-20 21:56:58	SUCCESS	\$14,500	Vendor A	N	N
6	8	2026-04-20 23:05:03	SUCCESS	\$14,200	Vendor A	N	N
7	8	2026-04-20 22:59:03	SUCCESS	\$13,500	Vendor C	N	N
8	8	2026-04-20 22:52:16	NO_AWARD	—	—	—	Y
9	8	2026-04-20 22:45:30	SUCCESS	\$13,500	Vendor C	N	N
10	8	2026-04-20 22:37:08	SUCCESS	\$14,200	Vendor A	N	N

3. Outcome Distributions

Measure	Scenario 4	Scenario 8
Status distribution	SUCCESS: 4 · NO_AWARD: 1	SUCCESS: 4 · NO_AWARD: 1
SUCCESS rate	4 / 5 (80%)	4 / 5 (80%)
Final price (SUCCESS)	\$13,500: 2 · \$14,500: 1 · \$15,000: 1	\$13,500: 2 · \$14,200: 2
Price range (SUCCESS)	\$13,500 – \$15,000	\$13,500 – \$14,200
Price spread (SUCCESS)	\$1,500 (~10% of ceiling)	\$700 (~5% of ceiling)
Winner (SUCCESS)	Vendor A: 2 · Vendor C: 2	Vendor A: 2 · Vendor C: 2
At exact \$15,000 ceiling	1 / 4 (25%)	0 / 4 (0%)
Pinned-harvest point value	\$14,750 (Vendor A)	\$15,000 (Vendor A)

Measure	Scenario 4	Scenario 8
Pinned value within re-run range?	Yes (\$13,500–\$15,000 in re-runs)	No (re-runs capped at \$14,200)

Scenario 8 is the clearest case in the study for reporting distributions rather than point values. The pinned-harvest value (\$15,000) does not fall within the observed re-run SUCCESS range (\$13,500–\$14,200). A reliability claim based on the pinned harvest alone would overstate ceiling pressure for this scenario under repeated sampling.

The governance chain held across every run; the specific outcomes inside it did not.

4. Limitations

- **Sample size.** $n = 5$ per scenario is the top of the range the Phase 2 feedback requested. Results are empirical evidence of outcome spread, not statistically significant estimates; a larger study ($n \geq 20$ per scenario) would support confidence intervals.
- **Coverage and conditions.** Only the borderline scenarios under Phase 3 conditions were re-run; all 10 runs were conducted on a single date with a single model version (claude-sonnet-4-6) and a single human operator.

Appendix A — Method

- **Scenarios and runs:** Scenarios 4 and 8, $n = 5$ each, total $n = 10$.
- **Date:** 2026-04-20, 21:56 UTC to 23:05 UTC.
- **Model:** claude-sonnet-4-6. Hard ceiling: \$15,000 (deterministic award_contract.py). Round cap: 3.
- **Data source:** logs/evidence_log_*.json only. The full 11-scenario harvest pipeline was not re-run; partial re-runs are sufficient for a distribution study.
- **Held constant:** scenario configuration, vendor personas, Advisor prompt, orchestrator logic, and guardrail. The variation across runs came from LLM sampling and human operator decisions.
- **Measured:** deal outcome (SUCCESS / NO_AWARD), final award price, winning vendor, at-ceiling flag, and a session-ended-no-contract flag (Y only when the session terminated without an executed contract — not when the human said “no” to an earlier proposal before approving a later one).
- **Traceability:** full run inventory with log filenames and session IDs available in evaluation/multi_seed_variance/runs.csv.